Automated Discovery of an Extremal Black Hole in a 4D Spacetime Simulation

QuantumVerse AI Agent
QuantumVerse Simulator, Autonomous Discovery Engine

LLM$_ABSTRACT$

INTRODUCTION

The QuantumVerse Simulator utilizes a PPO reinforcement learning agent to explore the parameter space of General Relativity. The agent navigates the black hole mass-spin-charge parameter space while respecting the cosmic censorship hypothesis, which forbids naked singularities. Each proposed state is validated by the PhysicsValidator module before acceptance.

METHODOLOGY

The RL agent explored the parameter space near the extremal limit ($a^2+Q^2 \approx M^2$). The proposed metric was validated against cosmic censorship using the PhysicsValidator module. The final validated parameters were determined to be:

- Mass (M): MASS $M_\odot$
- Spin (a): SPIN
- Charge (Q): CHARGE

The extremality parameter a^2+Q^2/M^2 was computed to verify proximity to the extremal bound without exceeding it.

RESULTS

The spacetime curvature was evaluated at the probe location using the Kerr metric curvature invariants. The Kretschmann scalar was found to be $K = KRETSCHMANN$. The gravitational redshift at the probe radius was calculated as $z = REDSHIFT$. All curvature invariants remained finite, confirming the absence of a naked singularity.

CONCLUSION

The AI agent successfully navigated the parameter space to find a stable, near-extremal black hole without violating cosmic censorship. The discovered spacetime satisfies all General Relativity validity checks, with curvature invariants well-behaved outside the event horizon.